

Transposable-element landscape and transcriptomic dysregulation across invasive mosquito vectors

University of Pavia - genomics RA project (>70 genomes analysed) · Gabriele Bambini · draft v1 · 2026-08-26

HYPOTHESIS

TE family expansion and TE-transcript activation differ systematically between vector species and population-invasive lineages, producing measurable immune-neighbourhood effects. A curated, per-species TE library is the missing substrate for reproducible TE-expression comparison at this scale.

AIMS

A1 consensus TE library per species (curation pass) | A2 TE-expression divergence maps across >70 genomes compendium | A3 immune-neighbourhood association + public browser release.

METHODS

Pipelines: RepeatModeler2 de novo families + EDTA-style curation; RepeatMasker genome masking; RNA-seq TE quantification (TEtranscripts approach) across the compendium; divergence tables per family superfamily; browser + library release. Compute: Pavia HPC (proven ImageJ/HPC track record extends naturally).

EXPECTED OUTPUTS

Curated TE libraries per species; divergence atlas; methods paper targeting bioinformatics venues; reusable Snakemake pipeline.

BIBLIOGRAPHY & CHAIN OF VERIFICATION (1/7 SOURCES LIVE ON 2026-08-26)

Claim used	Source	Link	CoV
Improved Aedes aegypti reference genome underpins reannotation	Matthews et al., Nature 2018	https://doi.org/10.1038/s41586-018-0692-z	Ⓢ unverified
TEtranscripts quantifies TE expression from RNA-seq	Jin et al., Bioinformatics 2015	https://doi.org/10.1093/bioinformatics/btv022	Ⓢ unverified
RepeatModeler2 de novo TE discovery pipeline	Flynn et al., 2020	https://www.biorxiv.org/content/10.1101/675722	Ⓢ unverified
RepeatMasker standard masking pipeline	Smit, Hubley & Green	https://www.repeatmasker.org/	Ⓢ unverified
VectorBase hosts vector genomes and annotations	VectorBase	https://vectorbase.org/	Ⓢ unverified

[CoV] VectorBase bulk downloads reachable	data source check	https://vectorbase.org/common/downloads	⊙ unverified
[CoV] NCBI Datasets usable for accession pulls	data source check	https://www.ncbi.nlm.nih.gov/datasets/	✓ HTTP 200

Chain-of-verification method: every claim row was probed automatically at build time (2026-08-26); ✓ = endpoint answered HTTP 200 during generation, ⊙ = probe failed or blocked (verify manually before citing in applications). Draft generated by the PhD Funding Command Center build.